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Electrical wire with path restricting member attached and wire harness

Abstract

An electrical wire structure including: an electrical wire; an exterior tube covering an outer periphery of the electrical wire; and a path restrictor covering a range spanning more than half of an outer periphery of the exterior tube in a circumferential direction of the exterior tube, extending along a lengthwise direction of the exterior tube, and restricting a path along which the electrical wire is routed.

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Background/Summary

BACKGROUND

(1) The present disclosure relates to an electrical wire with path restricting member attached and a wire harness.

(2) Conventionally, electrical wires with path restricting member attached that include a corrugated tube covering the outer periphery of an electrical wire member and a path restricting member covering part of the corrugated tube in the circumferential direction thereof and restricting the path along which the wire member is routed are known (e.g., see JP 2013-55760A).

(3) The corrugated tube of the electrical wire with path restricting member attached described in JP 2013-55760A has a lengthwise slit formed therein. The path restricting member includes a path maintaining member provided around the outer periphery of the corrugated tube and an attachment member provided within the slit. The attachment member is configured to be latchable onto both a portion on the inner peripheral side of the slit and a portion on the outer peripheral side of the path maintaining member. The path of the wire member is restricted due to the corrugated tube, the path maintaining member and the attachment member being fixed by taping or the like,

(4) A configuration is conceivable in which the wire harness is fixed to a vehicle body, due to the integrated corrugated tube and path restricting member being collectively bound together and held with a belt clamp, and the belt clamp being fixed to the vehicle body, for example.

SUMMARY

(5) Incidentally, since a path restricting member such as described above had a constant thickness, there is a problem that flexural rigidity is low. The flexural rigidity of the path restricting member being low is a factor leading to the wire member deviating from its path.

(6) An exemplary aspect of the disclosure provides an electrical wire with path restricting member attached and a wire harness that are capable of suppressing deviation of an electrical wire member from its path.

(7) An electrical wire with path restricting member attached of the present disclosure includes an electrical wire; an exterior tube covering an outer periphery of the electrical wire; and a path restrictor covering a range spanning more than half of an outer periphery of the exterior tube in a circumferential direction of the exterior tube, extending along a lengthwise direction of the exterior tube, and restricting a path along which the electrical wire is routed, wherein the path restrictor has: a main body covering a range spanning more than half of the outer periphery of the exterior tube in the circumferential direction of the exterior tube; an insertion opening formed by both circumferential ends of the main body, extending along a lengthwise direction of the path restrictor over an entirety thereof in the lengthwise direction, and configured for the exterior tube to be insertable therein; and a rib protruding from an outer periphery of the main body and extending along the lengthwise direction of the path restrictor.

(8) A wire harness of the present disclosure includes an electrical wire with a path restrictor attached; and a fixing member holding the electrical wire with the path restrictor attached, wherein the electrical wire with the path restrictor attached has: the electrical wire; an exterior tube covering an outer periphery of the electrical wire; and the path restrictor covering a range spanning more than half of an outer periphery of the exterior tube in a circumferential direction of the exterior tube, extending along a lengthwise direction of the exterior tube, and restricting a path along which the electrical wire is routed, the path restrictor has: a main body covering a range spanning more than half of the outer periphery of the exterior tube in the circumferential direction of the exterior tube; an insertion opening formed by both circumferential ends of the main body, extending along a lengthwise direction of the path restrictor over an entirety thereof in the lengthwise direction, and configured for the exterior tube to be insertable therein; and a rib protruding from an outer

periphery of the main body and extending along the lengthwise direction of the path restrictor, the main body has: a first constituent part forming one side of the main body with respect to a circumferential center of the main body; and a second constituent part forming the other side of the main body with respect to the circumferential center of the main body, the rib includes support ribs provided on both the first constituent part and the second constituent part, and the fixing member has a pair of supporting parts supporting the support ribs.
(9) According to an electrical wire with path restricting member attached and a wire harness of the present disclosure, deviation of an electrical wire member from its path can be suppressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic configuration diagram showing a wire harness of one embodiment.
(2) FIG. **2** is a cross-sectional view showing part of the wire harness of one embodiment.
(3) FIG. **3** is a partially exploded perspective view of an electrical wire with path restricting member attached of one embodiment.
(4) FIG. **4** is a side view in which the wire harness of one embodiment is fixed to a vehicle body.
DETAILED DESCRIPTION OF EMBODIMENTS
Description of Embodiments of Disclosure
(5) Initially, modes of the present disclosure will be enumerated and described.
(6) An electrical wire with path restricting member attached of the present disclosure includes:
(7) [1] an electrical wire member, a tubular exterior member coveting an outer periphery of the wire member, and a path restricting member covering a range spanning more than half of an outer periphery of the exterior member in a circumferential direction of the exterior member, extending along a lengthwise direction of the exterior member, and restricting a path along which the wire member is routed, the path restricting member having a main body part covering a range spanning more than half of the outer periphery of the exterior member in the circumferential direction of the exterior member, an insertion opening formed by both circumferential end portions of the main body part, extending along a lengthwise direction of the path restricting member over an entirety thereof in the lengthwise direction, and configured for the exterior member to be insertable therein, and a rib protruding from an outer periphery of the main body part and extending along the lengthwise direction of the path restricting member.
(8) According to this configuration, the path restricting member can be subsequently attached to the outer periphery of the exterior member through the insertion opening. Further, since the path restricting member has the rib that protrudes from the outer periphery of the main body part and extends along the lengthwise direction of the path restricting member, the flexural rigidity of the path restricting member can be increased. Therefore, deviation of the wire member from its path can be suppressed.
(9) [2] Preferably, the path restricting member is more rigid than the exterior member.
(10) According to this configuration, the path of the wire member can be more firmly restricted by the rigid path restricting member.
(11) [3] Preferably, the main body part has a first constituent part constituting one side of the main body part with respect to the circumferential center thereof, and a second constituent part constituting the other side of the main body part with respect to the circumferential center thereof, and the rib includes support ribs provided on both the first constituent part and the second constituent part.
(12) According to this configuration, since the rib includes the support ribs that are provided on both the first constituent part and second constituent part of the main body part, the electrical wire with path restricting member attached is easily supported by another member. That is, since the

support ribs are provided on the first constituent part constituting one side of the main body part with respect to the circumferential center thereof and the second constituent part constituting the other side of the main body part with respect to the circumferential center thereof, the electrical wire with path restricting member attached is easily held by the fixing part supporting the pair of support ribs.
(13) [4] Preferably, the rib includes guiding ribs provided on both circumferential end portions of the main body part, and spaced wider apart from each other proceeding in a direction in Which the ribs protrude.
(14) According to this configuration, since the rib includes the guiding ribs that are provided on both circumferential end portions of the main body part and are spaced wider apart from each other proceeding in the direction in which the ribs protrude, the exterior member can be smoothly inserted inside the path restricting member.
(15) A wire harness of the present disclosure includes:
(16) [5] an electrical wire with path restricting member attached and a fixing member holding the electrical wire with path restricting member attached, the electrical wire with path restricting member attached having an electrical wire member, a tubular exterior member covering an outer periphery of the wire member, and a path restricting member covering a range spanning more than half of an outer periphery of the exterior member in a circumferential direction of the exterior member, extending along a lengthwise direction of the exterior member, and restricting a path along which the wire member is routed, the path restricting member having a main body part covering a range spanning more than half of the outer periphery of the exterior member in the circumferential direction of the exterior member, an insertion opening formed by both circumferential end portions of the main body part, extending along a lengthwise direction of the path restricting member over an entirety thereof in the lengthwise direction, and configured for the exterior member to be insertable therein, and a rib protruding from an outer periphery of the main body part and extending along the lengthwise direction of the path restricting member, the main body part having a first constituent part constituting one side of the main body part with respect to the circumferential center thereof, and a second constituent part constituting the other side of the main body part with respect to the circumferential center thereof, the rib including support ribs provided on both the first constituent part and the second constituent part, and the fixing member having a pair of supporting parts supporting the support ribs.
(17) According to this configuration, the path restricting member can be subsequently attached to the outer periphery of the exterior member through the insertion opening. Further, since the path restricting member has the rib that protrudes from the outer periphery of the main body part and extends along the lengthwise direction of the path restricting member, the flexural rigidity of the path restricting member can be increased. Therefore, deviation of the wire member from its path can be suppressed. Also, since the rib includes the support ribs that are provided on both the first constituent part and the second constituent part of the main body part, the electrical wire with path restricting member attached is easily supported by another member. That is, since the support ribs are provided on the first constituent part constituting one side of the main body part with respect to the circumferential center thereof and the second constituent part constituting the other side of the main body part with respect to the circumferential center thereof, the electrical wire with path restricting member attached is easily held by the fixing part supporting the pair of support ribs. Further, the fixing member is easily able to hold the electrical wire with path restricting member attached due to supporting the pair of support ribs with the pair of supporting parts.
(18) [6] Preferably, the path restricting member is more rigid than the exterior member.
(19) According to this configuration, the path of the wire member can be more firmly restricted by the rigid path restricting member.
(20) [7] Preferably, the fixing member has an engaging part protruding toward the insertion opening, and the engaging part is contactable with at least one of the circumferential end portions

of the main body part,
(21) According to this configuration, since the engaging part of the fixing member is contactable with at least one of the two circumferential end portions of the main body part, the fixing member is able to suppress circumferential rotation of the path restricting member with respect to the fixing member while holding the path restricting member.
(22) [8] Preferably, the engaging part has a narrow portion narrowing in width proceeding toward a distal end side as seen from the lengthwise direction of the path restricting member.
(23) According to this configuration, since the engaging part has the narrow portion that narrows in width proceeding toward the distal end side as seen from the lengthwise direction of the path restricting member, the engaging part is easily inserted into the insertion opening, and assembly is facilitated.
(24) [9] Preferably, the rib includes guiding ribs provided on both circumferential end portions of the main body part, and spaced wider apart from each other proceeding in a direction in which the ribs protrude.
(25) According to this configuration, since the rib includes the guiding ribs that are provided on both circumferential end portions of the main body part and are spaced wider apart from each other proceeding in the direction in which the ribs protrude, the exterior member can be smoothly inserted inside the path restricting member.
(26) [10] Preferably, the rib includes guiding ribs provided on both circumferential end portions of the main body part, and spaced wider apart from each other proceeding in a direction in which the ribs protrude, and the engaging part is contactable with at least one of the guiding ribs.
(27) According to this configuration, since the rib includes the guiding ribs that are provided on both circumferential end portions of the main body part and are spaced wider apart from each other proceeding in the direction in which the ribs protrude, the exterior member can be smoothly inserted inside the path restricting member. Also, since the engaging part is contactable with at least one of the guiding ribs, circumferential rotation of the path restricting member with respect to the fixing member can be further suppressed.
(28) An electrical wire with path restricting member attached of the present disclosure is:
(29) [11] an electrical wire with path restricting member attached configured to be held in a fixing member having a pair of supporting parts, including an electrical wire member, a tubular exterior member covering an outer periphery of the wire member, and a path restricting member covering a range spanning more than half of an outer periphery of the exterior member in a circumferential direction of the exterior member, extending along a lengthwise direction of the exterior member, and restricting a path along which the wire member is routed, the path restricting member having a main body part covering a range spanning more than half of the outer periphery of the exterior member in the circumferential direction of the exterior member, an insertion opening formed by both circumferential al end portions of the main body part, extending along a lengthwise direction of the path restricting member over an entirety thereof in the lengthwise direction, and configured for the exterior member to be insertable therein, and a rib protruding from an outer periphery of the main body part and extending along the lengthwise direction of the path restricting member, the main body part having a first constituent part constituting one side of the main body part with respect to the circumferential center thereof, and a second constituent part constituting the other side of the main body part with respect to the circumferential center thereof, and the rib including support ribs provided on both the first constituent part and the second constituent part, and supported by the pair of supporting parts.
(30) According to this configuration, the path restricting member can be subsequently attached to the outer periphery of the exterior member through the insertion opening. Further, since the path restricting member has the rib that protrudes from the outer periphery of the main body part and extends along the lengthwise direction of the path restricting member, the flexural rigidity of the path restricting member can be increased. Therefore, deviation of the wire member from its path

can be suppressed. Moreover, since the rib includes the support ribs that are provided on both the first constituent part and the second constituent part of the main body part and are supported by the pair of supporting parts of the fixing member, the electrical wire with path restricting member attached is easily supported. That is, since the support ribs are provided on the first constituent part constituting one side of the main body part with respect to the circumferential center thereof and the second constituent part constituting the other side of the main body part with respect to the circumferential center thereof, the electrical wire with path restricting member attached is easily held by the pair of supporting parts supporting the support ribs.
(31) Preferably, the path restricting member is more rigid than the exterior member.
(32) According to this configuration, the path of the wire member can be more firmly restricted by the rigid path restricting member.
Detailed Description of Embodiments of Disclosure
(33) Specific examples of a wire harness of the present disclosure will be described below with reference to the drawings. In the individual diagrams, part of the configuration may be shown in an exaggerated manner or a simplified manner, for convenience of description. Also, the dimensional ratios of various portions may differ in the individual diagrams. Note that the present disclosure is not limited to these illustrative examples and is defined by the claims, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein. Herein, “orthogonal” is not only strictly orthogonal but also includes generally orthogonal within a range that achieves the operation and effect of the present embodiment. Also, herein, “circular” and “circular arc shaped” are not only strictly circular and circular arc shaped but also include generally circular and circular arc shaped within a range that achieves the operation and effect of the present embodiment.
Overall Configuration of Wire Harness **10**
(34) A wire harness **10** shown in FIG. **1** electrically connects two electrical devices or three or more electrical devices. The wire harness **10** electrically connects an inverter **11** installed in a front part of a vehicle V such as a hybrid vehicle or an electric vehicle, and a high voltage battery **12** installed in the vehicle V rearward of the inverter **11**. The wire harness **10** is, for example, routed so as to pass under the floor of the vehicle V. For example, an intermediate portion of the wire harness **10** in the lengthwise direction thereof is routed so as to pass outside the vehicle cabin such as under the floor of the vehicle V.
(35) The inverter **11** is connected to a motor for driving wheels (not shown) that serves as a power source for vehicle travel. The inverter **11** generates AC power from the DC power of the high voltage battery **12** and supplies the AC power to the motor. The high voltage battery **12** is, for example, a battery capable of supplying a voltage of several hundred volts.
(36) As shown in FIG. **2**, the wire harness **10** includes an electrical wire **13** with path restricting member attached. The electrical wire **13** with path restricting member attached includes a wire member **20** (electrical wire) that electrically connects the electrical devices to each other, a tubular exterior member **30** (exterior tube) that covers the outer periphery of the wire member **20**, and a path restricting member **40** (path restrictor) that covers the outer periphery of the exterior member **30** and restricts the path (hereinafter, routing path) along which the wire member **20** is routed.
(37) Also, as shown in FIGS. **2** and **4**, the wire harness **10** includes a fixing member **60** that holds the path restricting member **40** and is fixed to a vehicle body V**1** of the vehicle V. A pair of connectors C**1** and C**2** are respectively attached to either end portion of the wire member **20**.
(38) Configuration of Wire Member **20**
(39) The wire member **20** includes one or more electrical wires **21** and a braided member **24** that collectively covers the outer periphery of the one or more wires **21**, The wire member of the present embodiment has two wires **21**. One end portion of the wire member **20** is connected to the inverter **11** via the connector C**1**, and the other end portion of the wire member **20** is connected to the high voltage battery **12** via the connector C**2**. The wire member **20** is formed in an elongated

shape so as to extend in the front-rear direction of the vehicle, for example. The wires **21** are, for example, high voltage electrical wires capable of handling a high voltage and a large current. The wires **21** may, for example, be unshielded wires that do not have their own electromagnetic shielding structure, or shielded wires that have their own electromagnetic shielding structure.
(40) Configuration of Wire **21**
(41) As shown in FIG. **2**, the wire **21** is a coated electrical wire that has a core wire **22** composed of a conductor and an insulation coating **23** covering the outer periphery of the core wire **22**.
(42) Configuration of Core Wire **22**
(43) As the core wire **22**, a twisted wire formed by combining a plurality of metal wire strands, a columnar conductor composed of one columnar metal rod having a solid structure internally, or a tubular conductor having a hollow structure internally, for example, can be used. A plurality of types of conductors such as a twisted wire, a columnar conductor and a tubular conductor, for example, can also be used as the core wire **22**. Examples of the columnar conductor include a single core wire and a busbar. The core wire **22** of the present embodiment is a twisted wire. As the material of the core wire **22**, a copper-based or aluminum-based metal material, for example, can be used.
(44) The cross-sectional shape (hereinafter referred to as the transverse sectional shape) in which the core wire **22** is cut by a plane orthogonal to the lengthwise direction of the core wire **22**, that is, orthogonal to the lengthwise direction of the wire **21**, can be any shape. The transverse sectional shape of the core wire **22** is formed in a circular shape, a semicircular shape, a polygonal shape, a square shape or an oblate shape, for example. The transverse sectional shape of the core wire **22** of the present embodiment is formed in a circular shape.
(45) Configuration of Insulation Coating **23**
(46) The insulation coating **23** covers the outer peripheral surface of the core wire **22** around the entire circumference, for example. The insulation coating **23** is made of an insulating material such as a synthetic resin, for example. As the material of the insulation coating **23**, a synthetic resin whose main component is a polyolefin resin such as cross-linked polyethylene or cross-linked polypropylene, for example, can be used. Also, as the material of the insulation coating **23**, one type of material may be used alone, or two or more types of materials may be used in combination as appropriate.
(47) Configuration of Braided Member **24**
(48) The braided member **24** has a tubular shape that collectively covers the outer periphery of the wires **21** as a whole, for example. The braiding member **24** is provided so as to cover the outer periphery of the wires **21** along approximately the entirety of the wires **21** in the lengthwise direction thereof, for example. As the braided member **24**, a braided wire in which a plurality of metal wire strands are braided or a braided wire in which metal wire strands and resin wire strands are braided together can be used. As the material of the metal wire strands, a copper-based or aluminum-based metal material, for example, can be used. Although not shown in the diagrams, the braided member **24** is grounded at the connectors C**1** and C**2** and the like, for example.
(49) Configuration of Exterior Member **30**
(50) As shown in FIGS. **2** and **3**, the exterior member **30** has a cylindrical shape that covers the outer periphery of the wire member **20** around its entirety in the circumferential direction. The exterior member **30** is sealed around its entirety in the circumferential direction. The exterior member **30** is provided so as to cover a lengthwise part of the outer periphery of the wire member **20**, for example. As shown in FIG. **3**, the exterior member **30** of the present embodiment is a corrugated tube having a bellows structure in which an annular raised part **31** and an annular recessed part **32** are alternately provided continuously along the lengthwise direction thereof. The exterior member **30** has flexibility.
(51) As the material of the exterior member **30**, a resin material that has conductivity or a resin material that does not have conductivity, for example, can be used. As the resin material, a

synthetic resin such as polyolefin, polyimide, polyester or ABS resin, for example, can be used.
(52) Configuration of Path Restricting Member **40**
(53) As shown in FIGS. **2** and **4**, the path restricting member **40** covers a range spanning more than half of the outer periphery of the exterior member **30** in the circumferential direction of the exterior member **30** and extends along the lengthwise direction of the exterior member **30**. The path restricting member **40** of the present embodiment is attached to the outer periphery of a portion of the routing path of the wire member **20** where the exterior member **30** passing under the floor of the vehicle V or the like extends linearly. The path restricting member **40** is more rigid than the exterior member **30**. That is, the hardness of the path restricting member **40** is such that the path restricting member **40** does not easily bend in a direction orthogonal to the lengthwise direction of the wire harness **10**, as compared with the exterior member **30**.
(54) The path restricting member **40** is made of resin. As the material of the path restricting member **40**, a synthetic resin such as polypropylene, polyamide or polyacetal, for example, can be used. The path restricting member **40** can be manufactured by a known manufacturing method such as extrusion molding or injection molding, for example. The path restricting member **40** of the present embodiment has a constant cross-sectional shape as seen from the lengthwise direction. The path restricting member **40** is an extrusion molded article. Note that, as described above, the path restricting member **40** is more rigid than the exterior member **30**, but the constituent material of the path restricting member **40** need not be a material that is harder as a raw material than the constituent material of the exterior member **30**. That is, the constituent material of the path restricting member **40** may be a harder material or a softer material with respect to the constituent material of the exterior member **30**. Also, the constituent material of the path restricting member **40** may be the same material as the constituent material of the exterior member **30**.
(55) As shown in FIG. **2**, the path restricting member **40** has a main body part **41** (main body), an insertion opening **42**, and ribs **43**. The main body part **41** is formed so as to cover a range spanning more than half of the outer periphery of the exterior member **30** in the circumferential direction of the exterior member **30**. The insertion opening **42** is formed by both circumferential end portions **44** of the main body part **41**, extends along the lengthwise direction of the path restricting member **40** over the entirety thereof in the lengthwise direction, and is configured for the exterior member **30** to be insertable therein. The ribs **43** protrude from the outer periphery of the main body part **41** and extends along the lengthwise direction of the path restricting member **40**.
(56) Specifically, the main body part **41** includes a first constituent part **45** constituting one side of the main body part **41** with respect to the circumferential center thereof and a second constituent part **46** constituting the other side of the main body part **41** with respect to the circumferential center thereof. Further, the ribs **43** include support ribs **47** provided on both the first constituent part **45** and the second constituent part **46**. The pair of support ribs **47** protrude in directly opposite directions to each other. Also, the pair of support ribs **47** are set so as to protrude horizontally in a state where the path restricting member **40** is fixed to the vehicle body V**1**. Note that the insertion opening **42** is set so as to face upward in a state where the path restricting member **40** is fixed to the vehicle body V**1**.
(57) Also, the ribs **43** include guiding ribs **48** that are provided on both circumferential end portions **44** of the main body part **41**, and are spaced wider apart from each other proceeding in the direction in which the ribs **43** protrude.
(58) Also, the ribs **43** include a general rib **49** provided at the circumferential center of the main body part **41**, that is, at the boundary between the first constituent part **45** and the second constituent part **46**.
(59) The path restricting member **40** has a pair of protruding parts **50** that protrude toward the exterior member **30** inserted therein and contact the outer surface of the exterior member or, more specifically, the outer surface of the annular raised parts **31**. The protruding parts protrude from the inner surface of both circumferential end portions **44** of the main body part **41**. The transverse

sectional shape of the protruding parts **50** is a semicircular shape, for example. The protruding parts **50** extend along the lengthwise direction of the path restricting member **40** over the entirety thereof in the lengthwise direction.
(60) The insertion opening **42** extends over the entirety of the path restricting member **40** in the lengthwise direction thereof. The opening width of the insertion opening **42**, that is, the shortest distance between the both circumferential end portions **44** of the main body part **41** is smaller than the outer diameter of the exterior member **30**.
(61) When inserting the exterior member **30** into the first insertion opening **42** from a direction orthogonal to the lengthwise direction, the path restricting member **40** is elastically deformed and the opening width of the first insertion opening **42** is enlarged, Once the exterior member **30** is inserted inside the path restricting member **40**, the path restricting member **40** elastically returns toward its original shape. Since the opening width thereby becomes smaller than the outer diameter of the exterior member **30**, the path restricting member **40** is attached to the exterior member **30**.
(62) Configuration of Fixing Member **60**
(63) As shown in FIGS. **2** and **4**, the fixing member **60** holds the electrical wire **13** with path restricting member attached and is fixed to an attachment surface V**2** of the vehicle body V**1**. The attachment surface V**2** of the present embodiment is a surface under the floor of the vehicle V outside the vehicle cabin and opposes the ground. As shown in FIG. **4**, a plurality of fixing members **60** are provided in the lengthwise direction of the path restricting member **40**. In the present embodiment, two fixing members **60** are provided in the lengthwise direction of the path restricting member **40**.
(64) The fixing member **60** is made of resin. The fixing member **60** has a pair of supporting parts **61** that support the support ribs **47** of the path restricting member **40**.
(65) Specifically, as shown in FIG. **2**, the fixing member **60** includes a plate-like vehicle body opposing part **62** that opposes the attachment surface V**2** of the vehicle body V**1** and the supporting parts **61** that bend and extend from both widthwise sides of the vehicle body opposing part **62**.
(66) The vehicle body opposing part **62** has a fitting part **63** that is fitted into an attachment hole V**3** formed on the attachment surface V**2**. The supporting parts **61** are set such so as to extend downward in a state where the fixing member **60** is fixed to the vehicle body V**1**. Also, the pair of supporting parts **61** each have a supporting claw **64** that protrudes from the distal end of the supporting parts **61** on the sides facing each other. The supporting parts **61** support the support ribs **47** in a manner whereby the lower surface of the support ribs **47** is placed on the upper surface of the supporting claws **64**. In other words, an upper region of the electrical wire **13** with path restricting member attached is disposed between the pair of supporting parts **61**, and the electrical wire **13** with path restricting member attached is held in the fixing member **60** by the pair of support ribs **47** being supported on the pair of supporting claws **64**.
(67) The fixing member **60** has an engaging part **65** that protrudes toward the insertion opening **42** of the path restricting member **40** that is held. In other words, the fixing member **60** has the engaging part **65** that protrudes so as to be disposed inside the insertion opening **42** of the path restricting member **40** that is held. The engaging part **65** is contactable with both circumferential end portions **44** of the main body part **41** of the path restricting member **40**. The engaging part **65** is disposed between both circumferential end portions **44** of the main body part **41** of the path restricting member **40**, and, when the path restricting member **40** goes to rotate circumferentially with respect to the fixing member **60**, restricts rotation of the path restricting member **40** by contacting the end portions **44**.
(68) The engaging part **65** is provided on the vehicle body opposing part **62** of the fixing member **60**. Specifically, the engaging part **65** protrudes toward the insertion opening **42** from the vehicle body opposing part **62**. In other words, the engaging part **65** protrudes from the attachment surface V**2** side of the vehicle body V**1**. The engaging part **65** protrudes toward the ground. The insertion opening **42** of the path restricting member **40** thereby faces upward, which is the attachment

surface V**2** side.
(69) Also, the engaging part **65** has a narrow portion **66** that narrows in width proceeding toward the distal end side as seen from the lengthwise direction of the path restricting member **40**. In the present embodiment, the engaging part **65** is the narrow portion **66** that narrows in width proceeding toward the as the distal end side. Further, the engaging part **65** is contactable with both circumferential end portions **44** of the main body part **41** of the path restricting member **40**, and is contactable with the guiding ribs **48**. That is, the engaging part **65** is simultaneously contactable with both circumferential end portions **44** of the main body part **41** and the guiding ribs **48**. Note that the position of the path restricting member **40** with respect to the fixing member **60** is adjustable in the lengthwise direction of the path restricting member **40** while suppressing circumferential rotation of the path restricting member **40** with respect to the fixing member **60**.
(70) As shown in FIG. **4**, the wire harness **10** includes a slide restricting member **70** that restricts movement of the path restricting member **40** with respect to the exterior member **30** in the lengthwise direction of the exterior member **30**. The slide restricting member **70** of the present embodiment is an adhesive tape. The slide restricting member **70** restricts movement of the path restricting member **40** with respect to the exterior member **30**, by being wrapped from the path restricting member **40** to the exterior member **30** at both lengthwise ends of the path restricting member **40**.
(71) The operation of the present embodiment will now be described.
(72) According to the wire harness **10** of the present embodiment, the path restricting member **40** can be subsequently attached to the outer periphery of the exterior member **30** through the insertion opening **42**. In other words, the path restricting member **40** can be assembled to the exterior member **30** from a direction orthogonal to the lengthwise direction of the exterior member **30**. Since the path restricting member **40** has the pair of protruding parts that contact the outer surface of the exterior member **30**, detachment of the path restricting member **40** from the exterior member **30** through the insertion opening **42** can be suppressed.
(73) Further, since the path restricting member **40** has the ribs **43** that protrude from the outer periphery of the main body part **41** and extend along the lengthwise direction of the path restricting member **40**, the flexural rigidity of the path restricting member **40** can be increased. Therefore, for example, bending of the path restricting member **40** is suppressed even when subjected to vibration or an external force of some sort, and deviation of the exterior member and, consequently, the wire member **20**, from its path is suppressed.
(74) Also, the wire harness **10** can be fixed by holding the path restricting member **40** attached to the exterior member **30** in the fixing member **60** with the support ribs **47**, and fixing the fixing member **60** to the vehicle body V**1**. At this time, the engaging part **65** of the fixing member **60** is contactable with both circumferential end portions **44** of the main body part **41** of the path restricting member **40**, and thus circumferential rotation of the path restricting member **40** with respect to the fixing member **60** is suppressed.
(75) The effect of the present embodiment will now be described. (1) The path restricting member **40** can be subsequently attached to the outer periphery of the exterior member **30** through the insertion opening **42**. Further, since the path restricting member **40** has the ribs **43** that protrude from the outer periphery of the main body part **41** and extend along the lengthwise direction of the path restricting member **40**, the flexural rigidity of the path restricting member **40** can be increased. Therefore, deviation of the wire member **20** from its path can be suppressed. Because the flexural rigidity of the path restricting member **40** can be increased, it is also possible to reduce the thickness of the main body part **41**. (2) Since the path restricting member **40** is more rigid than the exterior member **30**, the path of the wire member **20** can be more firmly restricted by the rigid path restricting member **40**. (3) Since the ribs **43** include the support ribs **47** provided on both the first constituent part **45** and the second constituent part **46** of the main body part **41**, the electrical wire **13** with path restricting member attached is easily supported by another member. That is, the

support ribs **47** are provided on the first constituent part **45** constituting one side of the main body part **41** with respect to the circumferential center thereof and the second constituent part **46** constituting the other side of the main body part **41** with respect to the circumferential center thereof. Therefore, the electrical wire **13** with path restricting member attached is easily held by the fixing member **60** supporting the pair of support ribs **47** with the pair of support parts **61**. (4) Since the ribs **43** include the guiding ribs **48** that are provided on both circumferential end portions **44** of the main body part **41** and are spaced wider apart from each other proceeding in the direction in which the ribs **43** protrude, the exterior member **30** can be smoothly inserted inside the path restricting member **40**. (5) Since the fixing member **60** has the engaging part **65** and the engaging part **65** is contactable with both circumferential end portions **44** of the main body part **41**, the fixing member **60** is able to suppress rotation of the path restricting member **40** with respect to the fixing member **60** while holding the path restricting member **40**. Therefore, for example, rotation of the path restricting member **40** due to vibration or the like while the vehicle V is travelling is suppressed, and orientation of the insertion opening **42** downward such as opposing the ground is suppressed. As a result, for example, deterioration of the capability of the path restricting member **40** that protects the exterior member **30** from flying stones or the like is suppressed, and deterioration of the durability of the wire harness **10** can be suppressed. Also, since the fixing member **60** that is fixed to the vehicle body V**1** has the engaging part **65** that suppresses rotation of the path restricting member **40**, the number of components can be reduced as compared with the case where these functions are configured with different members. (6) Since the engaging part **65** has the narrow portion **66** that narrows in width proceeding toward the distal end side as seen from the lengthwise direction of the path restricting member **40**, the engaging part **65** is easily inserted into the insertion opening **42**, and assembly is facilitated. (7) Since the engaging part **65** is contactable with the guiding ribs **48**, circumferential rotation of the path restricting member **40** with respect to the fixing member **60** can be further suppressed.

Example Changes

(76) The present embodiment can be implemented in a changed manner as follows. The present embodiment and the following example changes can be implemented in combination with each other to the extent that there no technical inconsistencies. In the above embodiment, the ribs **43** include the support ribs **47** provided on both the first constituent part **45** and the second constituent part **46** of the main body part **41**, but the present disclosure is not limited thereto, and the path restricting member may be constituted to not have the support ribs **47**. Also, the pair of support ribs **47** protrude in directly opposite directions to each other, but may protrude in directions shifted from being directly opposite. Note that, in this case, the shape of the fixing member **60** together with the path restricting member **40** needs to be changed. In the above embodiment, the ribs **43** include the guiding ribs **48** that are provided on both circumferential end portions **44** of the main body part **41** and are spaced wider apart from each other proceeding in the direction in which the ribs **43** protrude, but the present disclosure is not limited thereto, and the path restricting member may be constituted to not have the guiding ribs **48**, In the above embodiment, the ribs **43** include the general rib **49** provided at the circumferential center of the main body part **41**, that is, at the boundary between the first constituent part **45** and the second constituent part **46**, but the present disclosure is not limited thereto, and the path restricting member may be constituted to not have the general rib **49**. The path restricting member may be constituted to have ribs other than the ribs **43** of the above embodiment, specifically, ribs other than the support ribs **47**, the guiding ribs **48**, and the general rib **49**. For example, in the path restricting member **40** of the above embodiment, ribs may be further provided between the support ribs **47** and the general rib **49**. In the above embodiment, the fixing member **60** has the engaging part **65**, and the engaging part **65** is contactable with both circumferential end portions **44** of the main body part **41**, but the present disclosure is not limited thereto, and the fixing member may be constituted to not have the engaging part **65**. Note that, in this case, circumferential rotation of the path restricting member **40**

with respect to the fixing member **60** is preferably suppressed with another configuration. In the above embodiment, the engaging part **65** has the narrow portion **66** that narrows in width proceeding toward the distal end side as seen from the lengthwise direction of the path restricting member **40**, but the present disclosure is not limited thereto, and the engaging part may be constituted to not have the narrow portion **66**. In the above embodiment, the engaging part **65** is contactable with the guiding ribs **48**, but is not limited thereto, and may be configured to not contact the guiding ribs **48**. In the above embodiment, the path restricting member **40** is more rigid than the exterior member **30**, but the present disclosure is not limited thereto, and the path restricting member **40** may be constituted to be of equivalent hardness to the exterior member **30** or softer than the exterior member **30**. That is, as long as the path restricting member **40** acts such that the exterior member **30** does not bend easily and is able to restrict the path along which the wire member **20** is routed, the path restricting member **40** may be more rigid than the exterior member **30**, may be softer than the exterior member **30**, or may be of equivalent hardness to the exterior member **30**. In the above embodiment, the path restricting member **40** is made of resin, but is not limited thereto, and may be made of metal, for example. The path restricting member **40** may, for example, be made of an iron-based, copper-based or aluminum-based metal material. Also, the path restricting member **40** may be constituted by a metal plate material, and the ribs **43** may be formed by bending the plate material. When the path restricting member **40** is made of metal in this way, an increase in the temperature inside the exterior member **30**, and, consequently, an increase in the temperature of the wire member **20**, can be suppressed, in cases such as when disposed at a position close to a heat source of the vehicle V, for example. Also, the ribs **43** need not be solid and may be hollow, regardless of whether the path restricting member **40** is made of resin or metal. In the above embodiment, two fixing members **60** are provided in the lengthwise direction of the path restricting member **40**, but the present disclosure is not limited thereto, and one fixing member or three or more fixing members may be provided in the lengthwise direction of the path restricting member **40**. In the above embodiment, the path restricting member **40** has the protruding parts **50** that contact the outer surface of the exterior member **30**, but is not limited thereto, and may be constituted to not have the protruding parts **50**. In the above embodiment, the fixing member **60** is fixed to the vehicle body V**1** by the fitting part **63**, but is not limited thereto, and may be fixed with another configuration. The exterior member **30** may have a metal layer including a metal material that is provided on the outer surface of the corrugated tube. Such a metal layer can be provided by plating, for example. The metal layer is preferably provided on the entire outer surface of the annular raised parts **31** and the annular recessed parts **32** of the corrugated tube. A metal material such as aluminum whose emissivity is small, for example, is preferably used for the innermost surface of the metal layer. According to such a configuration, an increase in the temperature inside the exterior member **30**, and, consequently, an increase in the temperature of the wire member **20**, can be suppressed, in cases such as when disposed at a position close to a heat source of the vehicle, for example. The exterior member **30** may have a slit that extends in the lengthwise direction of the exterior member **30**. In this case, the exterior member **30** is preferably sealed around the entirety thereof in the circumferential direction, by taping the outer periphery of the exterior member **30**, for example, so as to close the slit over the entirety thereof in the lengthwise direction of the exterior member **30**. Deterioration in the water stopping performance of the exterior member **30** having the slit can thereby be suppressed. The wire member **20** may have one wire **21** or may have three or more wires **21**. The braided member **24** can also be omitted from the wire member **20**. The wire harness **10** may include a plurality of path restricting members **40** that are provided at an interval from each other in the lengthwise direction of the exterior member **30**. The path restricting member **40** is not limited to a path restricting member provided under the floor of the vehicle V. As long as a portion of the routing path of the wire member extends linearly, the path restricting member **40** may, for example, be provided within the vehicle cabin of the vehicle V.

(77) As shown in FIG. **2**, the engaging part **65** of the fixing member **60** may have a shape that

narrows in width proceeding toward the distal end of the engaging part **65**, along both circumferential end portions **44** of the main body part **41** and the guiding ribs **48** of the path restricting member **40**, The entirety of the engaging part **65** that contacts or opposes both circumferential end portions **44** of the main body part **41** and the guiding ribs **48** of the path restricting member **40** may be the narrow portion **66**. The distal end of the engaging part **65**, which can be a radially inward-facing surface of the engaging part **65**, may have a shape that follows the outer periphery of the exterior member **30**. In the case where the outer periphery of the exterior member **30** has a convex shape, when seen from the lengthwise direction of the path restricting member **40** as shown in FIG. **2**, the distal end of the engaging part **65** may have a concave shape that corresponds to or matches the convex shape. The engaging part **65** may at least partially contact the exterior member **30**.

(78) As shown in FIG. **2**, the pair of supporting parts **61** of the fixing member **60** may contact the distal end of the corresponding support ribs **47**, so as to sandwich the electrical wire **13** with path restricting member attached.

(79) As shown in FIG. **2**, in some modes of the present disclosure, an electrical wire (**13**) with path restricting member attached can include a path restricting member (**40**) that is attached to the outer peripheral surface of the exterior member (**30**) by a snap-fit, for example, and is configured to restrict the lengthwise shape of the exterior member (**30**) and the wire member (**20**).

(80) As shown in FIGS. **2** and **3**, in some modes of the present disclosure, the radially inward-facing surface of the path restricting member (**40**) may have a non-protruding surface that can be a concave surface extending between both protruding parts (**50**).

(81) As shown in FIG. **2**, in some modes of the present disclosure, when the path restricting member (**40**) is attached to the exterior member (**30**), the path restricting member (**40**) may contact the radially outward-facing surface of the exterior member (**30**) at both protruding parts (**50**) of the path restricting member (**40**) and at part of the non-protruding surface of the path restricting member (**40**), a gap may be formed between other parts of the non-protruding surface of the path restricting member (**40**) excluding the part of the non-protruding surface that contacts the exterior member (**30**) and the radially outward-facing surface of the exterior member (**30**), and the gap may extend over the entire length of the path restricting member (**40**).

Claims

1. An electrical wire structure comprising: an electrical wire; an exterior tube covering an outer periphery of the electrical wire; and a path restrictor covering a range spanning more than half of an outer periphery of the exterior tube in a circumferential direction of the exterior tube, extending along a lengthwise direction of the exterior tube, and restricting a path along which the electrical wire is routed, wherein the path restrictor has: a main body covering a range spanning more than half of the outer periphery of the exterior tube in the circumferential direction of the exterior tube; an insertion opening formed by both circumferential ends of the main body, extending along a lengthwise direction of the path restrictor over an entirety thereof in the lengthwise direction, and configured for the exterior tube to be insertable therein; support ribs each protruding from an outer periphery of the main body except the circumferential ends thereof, and the support ribs extending along the lengthwise direction of the path restrictor; and guiding ribs provided on the circumferential ends of the main body to contact an engaging part of a fixing member fixed to a vehicle body, so as to suppress circumferential rotation of the path restrictor.
2. The electrical wire structure according to claim 1, wherein the path restrictor is more rigid than the exterior tube.
3. The electrical wire structure according to claim 1, wherein: the main body has: a first constituent part forming one side of the main body with respect to a circumferential center of the main body; and a second constituent part forming the other side of the main body with respect to the

circumferential center of the main body; and the support ribs are provided on both the first constituent part and the second constituent part.
4. The electrical wire structure according to claim 1, wherein the guiding ribs are spaced wider apart from each other proceeding in a direction in which the guiding ribs protrude.
5. A wire harness comprising: an electrical wire with a path restrictor attached; and a fixing member holding the electrical wire with the path restrictor attached, wherein the electrical wire with the path restrictor attached has: the electrical wire; an exterior tube covering an outer periphery of the electrical wire; and the path restrictor covering a range spanning more than half of an outer periphery of the exterior tube in a circumferential direction of the exterior tube, extending along a lengthwise direction of the exterior tube, and restricting a path along which the electrical wire is routed, the path restrictor has: a main body covering a range spanning more than half of the outer periphery of the exterior tube in the circumferential direction of the exterior tube; an insertion opening formed by both circumferential ends of the main body, extending along a lengthwise direction of the path restrictor over an entirety thereof in the lengthwise direction, and configured for the exterior tube to be insertable therein; support ribs each protruding from an outer periphery of the main body except the circumferential ends thereof, and the support ribs extending along the lengthwise direction of the path restrictor; and guiding ribs provided on the circumferential ends of the main body to contact an engaging part of the fixing member fixed to a vehicle body, so as to suppress circumferential rotation of the path restrictor, the main body has: a first constituent part forming one side of the main body with respect to a circumferential center of the main body; and a second constituent part forming the other side of the main body with respect to the circumferential center of the main body, the support ribs are provided on both the first constituent part and the second constituent part, and the fixing member has a pair of supporting parts supporting the support ribs.
6. The wire harness according to claim 5, wherein the path restrictor is more rigid than the exterior tube.
7. The wire harness according to claim 5, wherein: the fixing member has the engaging part protruding toward the insertion opening, and the engaging part is contactable with at least one of the circumferential ends of the main body.
8. The wire harness according to claim 7, wherein the engaging part has a narrow portion narrowing in width proceeding toward a distal end side as seen from the lengthwise direction of the path restrictor.
9. The wire harness according to claim 5, wherein the guiding ribs are spaced wider apart from each other proceeding in a direction in which the guiding ribs protrude.
10. The wire harness according to claim 7, wherein: the guiding ribs are spaced wider apart from each other proceeding in a direction in which the guiding ribs protrude, and the engaging part is contactable with of the guiding ribs.
11. An electrical wire structure configured to be held in a fixing member having a pair of supporting parts, comprising: an electrical wire; an exterior tube covering an outer periphery of the electrical wire; and a path restrictor covering a range spanning more than half of an outer periphery of the exterior tube in a circumferential direction of the exterior tube, extending along a lengthwise direction of the exterior tube, and restricting a path along which the electrical wire is routed, wherein the path restrictor has: a main body covering a range spanning more than half of the outer periphery of the exterior tube in the circumferential direction of the exterior tube; an insertion opening formed by both circumferential ends of the main body, extending along a lengthwise direction of the path restrictor over an entirety thereof in the lengthwise direction, and configured for the exterior tube to be insertable therein; support ribs each protruding from an outer periphery of the main body except the circumferential ends thereof, and the support ribs extending along the lengthwise direction of the path restrictor; and guiding ribs provided on the circumferential ends of the main body to contact an engaging part of the fixing member fixed to a vehicle body, so as to

suppress circumferential rotation of the path restrictor, the main body has: a first constituent part forming one side of the main body with respect to a circumferential center of the main body; and a second constituent part forming the other side of the main body with respect to the circumferential center of the main body; and the support ribs are provided on both the first constituent part and the second constituent part, and are configured to be supported by the pair of supporting parts.

12. The electrical wire structure according to claim 11, wherein the path restrictor is more rigid than the exterior tube.